

Problem 10

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Problem 1 Two runners start a race at the same time and finish in a tie. Prove that at some time during the race they have the same speed. (Hint: Consider the function $h(t) = f(t) - g(t)$, where $f(t)$ and $g(t)$ are the position of the first and second runner at time t , respectively.)

Explanation. Let $f(t)$ be the position of the first runner at time t , and let $g(t)$ be the position of the second runner at time t . Let T denote the time that the two runners finish the race (which is the same, since they finish in a tie). Also, let $h(t) = f(t) - g(t)$. Since $h(0) = 0$ and $h(T) = 0$, by Rolle's Theorem there exists some c with $0 < c < T$ such that $h'(c) = 0$. But since $h'(t) = f'(t) - g'(t)$, we have that $f'(c) = g'(c)$. Therefore, the runners have the same speed at time c .

